C-STGB Two-Lane Execution Lifecycle: Offline Training vs. Online Real-Time Inference

LANE 1: OFFLINE BATCH TRAINING ($\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{cal}}$)

1. Batch Ingest & Invariants

- Historical transaction log
- Kirchhoff flow & Hawkes rates

2. Latent GraphSMOTE

- Typology clustering & interp.
- Localized bilinear edge synthesis

3. Joint GNN-Tree Training

- Harmonic attention & edge gate
- Asymmetric Bayes risk loss

4. Conformal Calibration

- Non-conformity on $\mathcal{D}_{\text{cal}}$
- $\hat{q}^{(0)}, \hat{q}^{(1)} \rightarrow 1 - \alpha \geq 99\%$

Frozen Weights Θ^*, Bayes Cut τ^*, Quantiles $\hat{q}$ ⟶ Production Serving

LANE 2: ONLINE STREAMING FAST-PATH INFERENCE (<0.45 ms SLA)

1. Live Ingest & Top-15

- Zero-copy Arrow RecordBatch
- 15-hop dynamic ego-graph cache

2. Incremental Invariants

- Flow conservation Φ_{flow}
- Hawkes burst rate $\lambda_u(t)$

3. Frozen GNN & Bayes

- Forward pass on cached weights
- Risk score $\hat{p}_u \in [0, 1]$

4. 3-Tier CRC Triage

- Dual-quantile dispatch $\Gamma(u)$
- T1 Block / T2 Review / T3 STP